

MBColumn: Eurocode 2 PMM 500-Point Audit Report

Validation Scope: 5 Biaxial Angles x 100 Points per Angle

Core Configuration: $\alpha_{cc} = 0.80$, Enforced Rebar Yield at Pure Axial

Reference Benchmark: S-CONCRETE (EN 1992-1-1)

Angle 0° Detailed Comparison (100 Points)

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
0.0	0.0	753.5195	753.4783	-0.04	0.01%
-104.8	-104.8	781.9442	786.1941	4.25	0.54%
-209.5	-209.5	810.2543	818.2710	8.02	0.99%
-314.3	-314.3	838.3353	848.4165	10.08	1.20%
-419.1	-419.1	866.0726	875.4531	9.38	1.08%
-523.9	-523.9	893.3516	902.2693	8.92	1.00%
-628.6	-628.6	920.0577	927.2164	7.16	0.78%
-733.4	-733.4	946.0764	951.5458	5.47	0.58%
-838.2	-838.2	971.2931	975.5080	4.21	0.43%
-942.9	-942.9	995.5932	999.0229	3.43	0.34%
-1047.7	-1047.7	1018.8620	1022.0413	3.18	0.31%
-1152.5	-1152.5	1040.9850	1044.5529	3.57	0.34%
-1257.3	-1257.3	1061.8770	1066.4079	4.53	0.43%
-1362.0	-1362.0	1081.5690	1087.6749	6.11	0.56%
-1466.8	-1466.8	1100.1200	1108.3036	8.18	0.74%
-1571.6	-1571.6	1117.5890	1125.0634	7.47	0.67%
-1676.3	-1676.3	1134.0370	1140.2574	6.22	0.55%
-1781.1	-1781.1	1149.5230	1155.0373	5.51	0.48%
-1885.9	-1885.9	1164.1080	1169.4768	5.37	0.46%
-1990.7	-1990.7	1177.8500	1183.5723	5.72	0.49%
-2095.4	-2095.4	1190.8100	1197.2001	6.39	0.54%
-2200.2	-2200.2	1203.0470	1210.2935	7.25	0.60%
-2305.0	-2305.0	1214.6210	1223.0078	8.39	0.69%
-2409.7	-2409.7	1225.5660	1235.1458	9.58	0.78%
-2514.5	-2514.5	1235.8190	1246.6372	10.82	0.88%
-2619.3	-2619.3	1245.2880	1257.8236	12.54	1.01%
-2724.1	-2724.1	1253.8840	1265.1417	11.26	0.90%
-2828.8	-2828.8	1261.5160	1272.2428	10.73	0.85%
-2933.6	-2933.6	1268.0950	1277.3985	9.30	0.73%
-3038.4	-3038.4	1273.5310	1282.1913	8.66	0.68%
-3143.1	-3143.1	1277.7340	1286.5453	8.81	0.69%
-3247.9	-3247.9	1280.6140	1290.3203	9.71	0.76%
-3352.7	-3352.7	1282.0800	1294.0953	12.02	0.94%
-3457.5	-3457.5	1282.0430	1296.8066	14.76	1.15%
-3562.2	-3562.2	1280.4500	1299.2882	18.84	1.47%
-3667.0	-3667.0	1277.3910	1301.7698	24.38	1.91%
-3771.8	-3771.8	1272.9960	1299.3829	26.39	2.07%
-3876.5	-3876.5	1267.3910	1296.9092	29.52	2.33%
-3981.3	-3981.3	1260.7060	1294.4356	33.73	2.68%
-4086.1	-4086.1	1253.0680	1288.0585	34.99	2.79%
-4190.9	-4190.9	1244.6050	1278.2521	33.65	2.70%
-4295.6	-4295.6	1235.4460	1268.4459	33.00	2.67%
-4400.4	-4400.4	1225.7190	1258.6396	32.92	2.69%
-4505.2	-4505.2	1215.5510	1248.8334	33.28	2.74%
-4609.9	-4609.9	1205.0710	1237.9801	32.91	2.73%
-4714.7	-4714.7	1194.3800	1226.9663	32.59	2.73%
-4819.5	-4819.5	1183.4740	1215.9525	32.48	2.74%
-4924.3	-4924.3	1172.3190	1204.9385	32.62	2.78%
-5029.0	-5029.0	1160.8850	1193.9247	33.04	2.85%
-5133.8	-5133.8	1149.1390	1181.5725	32.43	2.82%
-5238.6	-5238.6	1137.0500	1168.7759	31.73	2.79%

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
-5343.3	-5343.3	1124.5860	1155.9794	31.39	2.79%
-5448.1	-5448.1	1111.7150	1143.1828	31.47	2.83%
-5552.9	-5552.9	1098.4050	1130.3862	31.98	2.91%
-5657.7	-5657.7	1084.6240	1117.3551	32.73	3.02%
-5762.4	-5762.4	1070.3410	1101.4975	31.16	2.91%
-5867.2	-5867.2	1055.5250	1085.6397	30.11	2.85%
-5972.0	-5972.0	1040.1510	1069.7820	29.63	2.85%
-6076.7	-6076.7	1024.1990	1053.9244	29.73	2.90%
-6181.5	-6181.5	1007.6440	1038.0666	30.42	3.02%
-6286.3	-6286.3	990.4652	1022.2090	31.74	3.20%
-6391.1	-6391.1	972.6389	1003.3433	30.70	3.16%
-6495.8	-6495.8	954.1429	983.6248	29.48	3.09%
-6600.6	-6600.6	934.9548	963.9064	28.95	3.10%
-6705.4	-6705.4	915.0518	944.1880	29.14	3.18%
-6810.1	-6810.1	894.4114	924.4694	30.06	3.36%
-6914.9	-6914.9	873.0113	904.7510	31.74	3.64%
-7019.7	-7019.7	850.8423	885.0326	34.19	4.02%
-7124.4	-7124.4	827.9501	862.5634	34.61	4.18%
-7229.2	-7229.2	804.3939	837.9357	33.54	4.17%
-7334.0	-7334.0	780.2328	813.3079	33.08	4.24%
-7438.8	-7438.8	755.5261	788.6800	33.15	4.39%
-7543.5	-7543.5	730.3330	764.0522	33.72	4.62%
-7648.3	-7648.3	704.7126	739.4245	34.71	4.93%
-7753.1	-7753.1	678.7243	714.7965	36.07	5.31%
-7857.8	-7857.8	652.4271	690.1688	37.74	5.78%
-7962.6	-7962.6	625.8802	665.0858	39.21	6.26%
-8067.4	-8067.4	599.1430	635.0177	35.87	5.99%
-8172.2	-8172.2	572.2672	604.9452	32.68	5.71%
-8276.9	-8276.9	545.2750	574.8680	29.59	5.43%
-8381.7	-8381.7	518.1813	544.7858	26.60	5.13%
-8486.5	-8486.5	491.0013	514.6978	23.70	4.83%
-8591.2	-8591.2	463.7496	484.6040	20.85	4.50%
-8696.0	-8696.0	436.4413	454.5032	18.06	4.14%
-8800.8	-8800.8	409.0912	424.3955	15.30	3.74%
-8905.6	-8905.6	381.7142	394.2796	12.57	3.29%
-9010.3	-9010.3	354.3254	364.1547	9.83	2.77%
-9115.1	-9115.1	326.9395	334.0198	7.08	2.17%
-9219.9	-9219.9	299.5715	303.8733	4.30	1.44%
-9324.6	-9324.6	272.2333	273.7121	1.48	0.54%
-9429.4	-9429.4	244.9250	243.5356	-1.39	0.57%
-9534.2	-9534.2	217.6435	213.3421	-4.30	1.98%
-9639.0	-9639.0	190.3859	183.1229	-7.26	3.81%
-9743.7	-9743.7	163.1491	152.7419	-10.41	6.38%
-9848.5	-9848.5	135.9303	120.9636	-14.97	11.01%
-9953.3	-9953.3	108.7263	88.5355	-20.19	18.57%
-10058.0	-10058.0	81.5343	52.8812	-28.65	35.14%
-10162.8	-10162.8	54.3512	8.4584	-45.89	84.44%
-10267.6	-10267.6	27.1741	1.5810	-25.59	94.18%
-10372.4	-10372.4	0.0000	0.0000	0.00	0.00%

Angle 0° Audit Summary:
Average Deviation: 4.69% | Maximum Divergence: 94.18%

Angle 16° Detailed Comparison (100 Points)

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
0.0	0.0	724.4139	732.6795	8.27	1.14%
-104.8	-104.8	747.9668	762.3487	14.38	1.92%
-209.5	-209.5	771.4236	790.1320	18.71	2.43%
-314.3	-314.3	794.6880	817.2392	22.55	2.84%
-419.1	-419.1	817.6638	843.0448	25.38	3.10%
-523.9	-523.9	840.2549	867.4365	27.18	3.23%
-628.6	-628.6	862.3651	891.5536	29.19	3.38%
-733.4	-733.4	883.8981	915.3643	31.47	3.56%
-838.2	-838.2	904.7578	938.7059	33.95	3.75%
-942.9	-942.9	924.8480	961.4724	36.62	3.96%
-1047.7	-1047.7	944.0726	983.7695	39.70	4.20%
-1152.5	-1152.5	962.3352	1005.6876	43.35	4.50%
-1257.3	-1257.3	979.5587	1026.9637	47.41	4.84%
-1362.0	-1362.0	995.7408	1047.3142	51.57	5.18%
-1466.8	-1466.8	1010.8980	1067.0161	56.12	5.55%
-1571.6	-1571.6	1025.0480	1084.6844	59.64	5.82%
-1676.3	-1676.3	1038.2060	1101.5755	63.37	6.10%
-1781.1	-1781.1	1050.3910	1116.1380	65.75	6.26%
-1885.9	-1885.9	1061.6170	1130.4881	68.87	6.49%
-1990.7	-1990.7	1071.9020	1143.9909	72.09	6.73%
-2095.4	-2095.4	1081.2640	1157.4399	76.18	7.05%
-2200.2	-2200.2	1089.7180	1169.8554	80.14	7.35%
-2305.0	-2305.0	1097.2810	1182.2708	84.99	7.75%
-2409.7	-2409.7	1103.9710	1193.8174	89.85	8.14%
-2514.5	-2514.5	1109.8100	1205.0901	95.28	8.59%
-2619.3	-2619.3	1114.8210	1215.4629	100.64	9.03%
-2724.1	-2724.1	1119.0280	1223.7259	104.70	9.36%
-2828.8	-2828.8	1122.4530	1231.9888	109.54	9.76%
-2933.6	-2933.6	1125.1200	1237.8407	112.72	10.02%
-3038.4	-3038.4	1127.0520	1241.0792	114.03	10.12%
-3143.1	-3143.1	1128.2710	1244.3176	116.05	10.29%
-3247.9	-3247.9	1128.8020	1246.9230	118.12	10.46%
-3352.7	-3352.7	1128.6660	1246.0667	117.40	10.40%
-3457.5	-3457.5	1127.8880	1245.2103	117.32	10.40%
-3562.2	-3562.2	1126.4900	1244.3540	117.86	10.46%
-3667.0	-3667.0	1124.4950	1242.3237	117.83	10.48%
-3771.8	-3771.8	1121.9270	1237.1821	115.26	10.27%
-3876.5	-3876.5	1118.8100	1232.0405	113.23	10.12%
-3981.3	-3981.3	1115.1660	1226.8990	111.73	10.02%
-4086.1	-4086.1	1111.0190	1221.7574	110.74	9.97%
-4190.9	-4190.9	1106.3920	1214.6550	108.26	9.79%
-4295.6	-4295.6	1101.3080	1206.5350	105.23	9.55%
-4400.4	-4400.4	1095.7900	1198.4150	102.62	9.37%
-4505.2	-4505.2	1089.8630	1190.2950	100.43	9.22%
-4609.9	-4609.9	1083.5480	1182.1750	98.63	9.10%
-4714.7	-4714.7	1076.8640	1173.0594	96.20	8.93%
-4819.5	-4819.5	1069.8050	1162.9490	93.14	8.71%
-4924.3	-4924.3	1062.3590	1152.8384	90.48	8.52%
-5029.0	-5029.0	1054.5140	1142.7279	88.21	8.37%
-5133.8	-5133.8	1046.2590	1132.6174	86.36	8.25%
-5238.6	-5238.6	1037.5830	1122.5068	84.92	8.18%

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
-5343.3	-5343.3	1028.4720	1109.9764	81.50	7.92%
-5448.1	-5448.1	1018.9170	1096.8718	77.95	7.65%
-5552.9	-5552.9	1008.9040	1083.7671	74.86	7.42%
-5657.7	-5657.7	998.4232	1070.6625	72.24	7.24%
-5762.4	-5762.4	987.4620	1057.5580	70.10	7.10%
-5867.2	-5867.2	976.0056	1044.4533	68.45	7.01%
-5972.0	-5972.0	964.0253	1029.6090	65.58	6.80%
-6076.7	-6076.7	951.4895	1012.9014	61.41	6.45%
-6181.5	-6181.5	938.3661	996.1935	57.83	6.16%
-6286.3	-6286.3	924.6234	979.4858	54.86	5.93%
-6391.1	-6391.1	910.2293	962.7782	52.55	5.77%
-6495.8	-6495.8	895.1521	946.0703	50.92	5.69%
-6600.6	-6600.6	879.3597	929.3627	50.00	5.69%
-6705.4	-6705.4	862.8204	911.3615	48.54	5.63%
-6810.1	-6810.1	845.5023	890.0856	44.58	5.27%
-6914.9	-6914.9	827.3734	868.8100	41.44	5.01%
-7019.7	-7019.7	808.4139	847.5343	39.12	4.84%
-7124.4	-7124.4	788.6516	826.2584	37.61	4.77%
-7229.2	-7229.2	768.1265	804.9827	36.86	4.80%
-7334.0	-7334.0	746.8784	783.7071	36.83	4.93%
-7438.8	-7438.8	724.9474	762.4312	37.48	5.17%
-7543.5	-7543.5	702.3732	740.9407	38.57	5.49%
-7648.3	-7648.3	679.1957	715.6673	36.47	5.37%
-7753.1	-7753.1	655.4550	690.3935	34.94	5.33%
-7857.8	-7857.8	631.1908	665.1200	33.93	5.38%
-7962.6	-7962.6	606.4431	639.8465	33.40	5.51%
-8067.4	-8067.4	581.2517	614.5728	33.32	5.73%
-8172.2	-8172.2	555.6584	589.2993	33.64	6.05%
-8276.9	-8276.9	529.7117	564.0258	34.31	6.48%
-8381.7	-8381.7	503.4622	538.7523	35.29	7.01%
-8486.5	-8486.5	476.9602	510.5084	33.55	7.03%
-8591.2	-8591.2	450.2561	480.4681	30.21	6.71%
-8696.0	-8696.0	423.4004	450.4166	27.02	6.38%
-8800.8	-8800.8	396.4435	420.3523	23.91	6.03%
-8905.6	-8905.6	369.4357	390.2740	20.84	5.64%
-9010.3	-9010.3	342.4276	360.1809	17.75	5.18%
-9115.1	-9115.1	315.4695	330.0716	14.60	4.63%
-9219.9	-9219.9	288.6118	299.9402	11.33	3.93%
-9324.6	-9324.6	261.8943	269.7876	7.89	3.01%
-9429.4	-9429.4	235.3140	239.6070	4.29	1.82%
-9534.2	-9534.2	208.8571	209.3966	0.54	0.26%
-9639.0	-9639.0	182.5100	179.1436	-3.37	1.84%
-9743.7	-9743.7	156.2588	148.8412	-7.42	4.75%
-9848.5	-9848.5	130.0900	118.4726	-11.62	8.93%
-9953.3	-9953.3	103.9897	87.3778	-16.61	15.97%
-10058.0	-10058.0	77.9443	47.0941	-30.85	39.58%
-10162.8	-10162.8	51.9400	7.0239	-44.92	86.48%
-10267.6	-10267.6	25.9631	1.3153	-24.65	94.93%
-10372.4	-10372.4	0.0000	0.0000	0.00	0.00%

Angle 16° Audit Summary:
Average Deviation: 8.56% | Maximum Divergence: 94.93%

Angle 75° Detailed Comparison (100 Points)

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
0.0	0.0	549.5673	584.0640	34.50	6.28%
-104.8	-104.8	569.3040	603.4730	34.17	6.00%
-209.5	-209.5	588.9707	621.3909	32.42	5.50%
-314.3	-314.3	608.4974	638.1541	29.66	4.87%
-419.1	-419.1	627.8140	654.4356	26.62	4.24%
-523.9	-523.9	646.8505	670.5136	23.66	3.66%
-628.6	-628.6	665.5369	686.4556	20.92	3.14%
-733.4	-733.4	683.8032	702.2080	18.40	2.69%
-838.2	-838.2	701.5793	716.9651	15.39	2.19%
-942.9	-942.9	718.7953	731.5348	12.74	1.77%
-1047.7	-1047.7	735.3811	745.7753	10.39	1.41%
-1152.5	-1152.5	751.2667	759.7486	8.48	1.13%
-1257.3	-1257.3	766.3929	773.4674	7.07	0.92%
-1362.0	-1362.0	780.7441	786.0735	5.33	0.68%
-1466.8	-1466.8	794.3157	797.7270	3.41	0.43%
-1571.6	-1571.6	807.1029	808.9867	1.88	0.23%
-1676.3	-1676.3	819.1009	819.3891	0.29	0.04%
-1781.1	-1781.1	830.3050	829.7915	-0.51	0.06%
-1885.9	-1885.9	840.7104	839.0207	-1.69	0.20%
-1990.7	-1990.7	850.3126	848.1646	-2.15	0.25%
-2095.4	-2095.4	859.1066	856.8928	-2.21	0.26%
-2200.2	-2200.2	867.0878	864.6634	-2.42	0.28%
-2305.0	-2305.0	874.2515	872.4338	-1.82	0.21%
-2409.7	-2409.7	880.5966	879.1126	-1.48	0.17%
-2514.5	-2514.5	886.1371	883.9451	-2.19	0.25%
-2619.3	-2619.3	890.8908	888.7776	-2.11	0.24%
-2724.1	-2724.1	894.8754	893.6100	-1.27	0.14%
-2828.8	-2828.8	898.1086	895.9354	-2.17	0.24%
-2933.6	-2933.6	900.6081	898.1116	-2.50	0.28%
-3038.4	-3038.4	902.3916	900.2877	-2.10	0.23%
-3143.1	-3143.1	903.4769	902.4638	-1.01	0.11%
-3247.9	-3247.9	903.8816	902.4656	-1.42	0.16%
-3352.7	-3352.7	903.6235	902.4498	-1.17	0.13%
-3457.5	-3457.5	902.7204	902.4340	-0.29	0.03%
-3562.2	-3562.2	901.1931	902.4183	1.23	0.14%
-3667.0	-3667.0	899.0752	900.8925	1.82	0.20%
-3771.8	-3771.8	896.4036	898.0085	1.60	0.18%
-3876.5	-3876.5	893.2151	895.1244	1.91	0.21%
-3981.3	-3981.3	889.5464	892.2403	2.69	0.30%
-4086.1	-4086.1	885.4346	889.3562	3.92	0.44%
-4190.9	-4190.9	880.9162	885.5452	4.63	0.53%
-4295.6	-4295.6	876.0282	880.7652	4.74	0.54%
-4400.4	-4400.4	870.8074	875.9852	5.18	0.59%
-4505.2	-4505.2	865.2906	871.2052	5.91	0.68%
-4609.9	-4609.9	859.5146	866.4252	6.91	0.80%
-4714.7	-4714.7	853.5068	861.6453	8.14	0.95%
-4819.5	-4819.5	847.2570	855.1875	7.93	0.94%
-4924.3	-4924.3	840.7457	848.4634	7.72	0.92%
-5029.0	-5029.0	833.9531	841.7394	7.79	0.93%
-5133.8	-5133.8	826.8598	835.0154	8.16	0.99%
-5238.6	-5238.6	819.4460	828.2913	8.85	1.08%

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
-5343.3	-5343.3	811.6923	821.5673	9.87	1.22%
-5448.1	-5448.1	803.5789	813.1201	9.54	1.19%
-5552.9	-5552.9	795.0863	803.8659	8.78	1.10%
-5657.7	-5657.7	786.1949	794.6118	8.42	1.07%
-5762.4	-5762.4	776.8850	785.3577	8.47	1.09%
-5867.2	-5867.2	767.1378	776.1036	8.97	1.17%
-5972.0	-5972.0	756.9376	766.8495	9.91	1.31%
-6076.7	-6076.7	746.2690	757.5954	11.33	1.52%
-6181.5	-6181.5	735.1173	745.9783	10.86	1.48%
-6286.3	-6286.3	723.4672	733.7731	10.31	1.42%
-6391.1	-6391.1	711.3036	721.5679	10.26	1.44%
-6495.8	-6495.8	698.6116	709.3626	10.75	1.54%
-6600.6	-6600.6	685.3760	697.1575	11.78	1.72%
-6705.4	-6705.4	671.5817	684.9523	13.37	1.99%
-6810.1	-6810.1	657.2137	672.7470	15.53	2.36%
-6914.9	-6914.9	642.2569	660.3232	18.07	2.81%
-7019.7	-7019.7	626.7050	644.5112	17.81	2.84%
-7124.4	-7124.4	610.5862	628.6990	18.11	2.97%
-7229.2	-7229.2	593.9380	612.8870	18.95	3.19%
-7334.0	-7334.0	576.7975	597.0750	20.28	3.52%
-7438.8	-7438.8	559.2019	581.2629	22.06	3.95%
-7543.5	-7543.5	541.1882	565.4509	24.26	4.48%
-7648.3	-7648.3	522.7938	549.6388	26.85	5.13%
-7753.1	-7753.1	504.0558	533.8267	29.77	5.91%
-7857.8	-7857.8	485.0114	515.0722	30.06	6.20%
-7962.6	-7962.6	465.6978	495.6296	29.93	6.43%
-8067.4	-8067.4	446.1521	476.1868	30.03	6.73%
-8172.2	-8172.2	426.4090	456.7442	30.34	7.11%
-8276.9	-8276.9	406.4931	437.3016	30.81	7.58%
-8381.7	-8381.7	386.4265	417.8589	31.43	8.13%
-8486.5	-8486.5	366.2312	398.4161	32.18	8.79%
-8591.2	-8591.2	345.9000	378.9735	33.07	9.56%
-8696.0	-8696.0	325.5427	355.5193	29.98	9.21%
-8800.8	-8800.8	305.0937	331.4007	26.31	8.62%
-8905.6	-8905.6	284.6042	307.2869	22.68	7.97%
-9010.3	-9010.3	264.0964	283.1784	19.08	7.23%
-9115.1	-9115.1	243.5922	259.0771	15.48	6.36%
-9219.9	-9219.9	223.1137	234.9825	11.87	5.32%
-9324.6	-9324.6	202.6786	210.8972	8.22	4.05%
-9429.4	-9429.4	182.2869	186.8214	4.53	2.49%
-9534.2	-9534.2	161.9344	162.7593	0.82	0.51%
-9639.0	-9639.0	141.6166	138.7125	-2.90	2.05%
-9743.7	-9743.7	121.3291	114.6150	-6.71	5.53%
-9848.5	-9848.5	101.0678	90.0909	-10.98	10.86%
-9953.3	-9953.3	80.8282	65.1465	-15.68	19.40%
-10058.0	-10058.0	60.6059	33.0249	-27.58	45.51%
-10162.8	-10162.8	40.3967	4.7133	-35.68	88.33%
-10267.6	-10267.6	20.1962	0.8817	-19.31	95.63%
-10372.4	-10372.4	0.0000	0.0000	0.00	0.00%

Angle 75° Audit Summary:
Average Deviation: 4.95% | Maximum Divergence: 95.63%

Angle 90° Detailed Comparison (100 Points)

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
0.0	0.0	545.0552	545.1081	0.05	0.01%
-104.8	-104.8	566.2994	568.5818	2.28	0.40%
-209.5	-209.5	587.4559	591.2595	3.80	0.65%
-314.3	-314.3	608.4371	611.3647	2.93	0.48%
-419.1	-419.1	629.1551	631.0986	1.94	0.31%
-523.9	-523.9	649.5223	650.6487	1.13	0.17%
-628.6	-628.6	669.4511	670.0060	0.55	0.08%
-733.4	-733.4	688.8537	689.1345	0.28	0.04%
-838.2	-838.2	707.6424	708.0011	0.36	0.05%
-942.9	-942.9	725.7296	726.5748	0.85	0.12%
-1047.7	-1047.7	743.0274	744.8502	1.82	0.25%
-1152.5	-1152.5	759.4483	762.2890	2.84	0.37%
-1257.3	-1257.3	774.9274	778.4199	3.49	0.45%
-1362.0	-1362.0	789.4908	794.1136	4.62	0.59%
-1466.8	-1466.8	803.1879	809.0071	5.82	0.72%
-1571.6	-1571.6	816.0677	821.2735	5.21	0.64%
-1676.3	-1676.3	828.1792	832.4952	4.32	0.52%
-1781.1	-1781.1	839.5717	843.4565	3.88	0.46%
-1885.9	-1885.9	850.2942	854.1592	3.86	0.45%
-1990.7	-1990.7	860.3958	864.5995	4.20	0.49%
-2095.4	-2095.4	869.9257	874.7525	4.83	0.55%
-2200.2	-2200.2	878.9329	884.4418	5.51	0.63%
-2305.0	-2305.0	887.4667	893.8404	6.37	0.72%
-2409.7	-2409.7	895.5552	902.8878	7.33	0.82%
-2514.5	-2514.5	903.1438	911.3676	8.22	0.91%
-2619.3	-2619.3	910.1569	919.4580	9.30	1.02%
-2724.1	-2724.1	916.5189	923.9022	7.38	0.81%
-2828.8	-2828.8	922.1542	928.3464	6.19	0.67%
-2933.6	-2933.6	926.9873	932.1901	5.20	0.56%
-3038.4	-3038.4	930.9427	935.9187	4.98	0.53%
-3143.1	-3143.1	933.9447	939.3159	5.37	0.58%
-3247.9	-3247.9	935.9178	942.2708	6.35	0.68%
-3352.7	-3352.7	936.7864	945.2256	8.44	0.90%
-3457.5	-3457.5	936.4750	947.3765	10.90	1.16%
-3562.2	-3562.2	934.9398	949.3510	14.41	1.54%
-3667.0	-3667.0	932.2645	951.3256	19.06	2.04%
-3771.8	-3771.8	928.5644	947.1045	18.54	2.00%
-3876.5	-3876.5	923.9550	942.7627	18.81	2.04%
-3981.3	-3981.3	918.5518	938.4209	19.87	2.16%
-4086.1	-4086.1	912.4700	933.8849	21.41	2.35%
-4190.9	-4190.9	905.8252	927.1149	21.29	2.35%
-4295.6	-4295.6	898.7328	920.3450	21.61	2.40%
-4400.4	-4400.4	891.3082	913.5749	22.27	2.50%
-4505.2	-4505.2	883.6669	906.8050	23.14	2.62%
-4609.9	-4609.9	875.9243	899.6048	23.68	2.70%
-4714.7	-4714.7	868.1694	891.8718	23.70	2.73%
-4819.5	-4819.5	860.3869	884.1387	23.75	2.76%
-4924.3	-4924.3	852.5347	876.4056	23.87	2.80%
-5029.0	-5029.0	844.5709	868.6725	24.10	2.85%
-5133.8	-5133.8	836.4537	860.4346	23.98	2.87%
-5238.6	-5238.6	828.1412	851.3080	23.17	2.80%

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
-5343.3	-5343.3	819.5916	842.1816	22.59	2.76%
-5448.1	-5448.1	810.7629	833.0551	22.29	2.75%
-5552.9	-5552.9	801.6134	823.9286	22.32	2.78%
-5657.7	-5657.7	792.1010	814.8022	22.70	2.87%
-5762.4	-5762.4	782.1840	804.1520	21.97	2.81%
-5867.2	-5867.2	771.8251	792.9250	21.10	2.73%
-5972.0	-5972.0	761.0066	781.6981	20.69	2.72%
-6076.7	-6076.7	749.7151	770.4712	20.76	2.77%
-6181.5	-6181.5	737.9375	759.2442	21.31	2.89%
-6286.3	-6286.3	725.6606	748.0173	22.36	3.08%
-6391.1	-6391.1	712.8713	735.5224	22.65	3.18%
-6495.8	-6495.8	699.5562	721.0692	21.51	3.08%
-6600.6	-6600.6	685.7022	706.6161	20.91	3.05%
-6705.4	-6705.4	671.2962	692.1630	20.87	3.11%
-6810.1	-6810.1	656.3250	677.7098	21.38	3.26%
-6914.9	-6914.9	640.7752	663.2567	22.48	3.51%
-7019.7	-7019.7	624.6434	648.8036	24.16	3.87%
-7124.4	-7124.4	607.9642	633.5611	25.60	4.21%
-7229.2	-7229.2	590.7814	615.3642	24.58	4.16%
-7334.0	-7334.0	573.1395	597.1674	24.03	4.19%
-7438.8	-7438.8	555.0825	578.9704	23.89	4.30%
-7543.5	-7543.5	536.6544	560.7735	24.12	4.49%
-7648.3	-7648.3	517.8995	542.5766	24.68	4.76%
-7753.1	-7753.1	498.8618	524.3796	25.52	5.12%
-7857.8	-7857.8	479.5855	506.1827	26.60	5.55%
-7962.6	-7962.6	460.1148	487.9859	27.87	6.06%
-8067.4	-8067.4	440.4937	466.5416	26.05	5.91%
-8172.2	-8172.2	420.7609	444.2365	23.48	5.58%
-8276.9	-8276.9	400.9341	421.9346	21.00	5.24%
-8381.7	-8381.7	381.0251	399.6363	18.61	4.88%
-8486.5	-8486.5	361.0461	377.3417	16.30	4.51%
-8591.2	-8591.2	341.0092	355.0516	14.04	4.12%
-8696.0	-8696.0	320.9264	332.7659	11.84	3.69%
-8800.8	-8800.8	300.8098	310.4857	9.68	3.22%
-8905.6	-8905.6	280.6715	288.2112	7.54	2.69%
-9010.3	-9010.3	260.5235	265.9431	5.42	2.08%
-9115.1	-9115.1	240.3779	243.6822	3.30	1.37%
-9219.9	-9219.9	220.2468	221.4290	1.18	0.54%
-9324.6	-9324.6	200.1399	199.1853	-0.95	0.48%
-9429.4	-9429.4	180.0571	176.9535	-3.10	1.72%
-9534.2	-9534.2	159.9960	154.7345	-5.26	3.29%
-9639.0	-9639.0	139.9543	132.5320	-7.42	5.30%
-9743.7	-9743.7	119.9294	109.9349	-9.99	8.33%
-9848.5	-9848.5	99.9190	86.5811	-13.34	13.35%
-9953.3	-9953.3	79.9207	63.0875	-16.83	21.06%
-10058.0	-10058.0	59.9321	35.7652	-24.17	40.32%
-10162.8	-10162.8	39.9507	5.6491	-34.30	85.86%
-10267.6	-10267.6	19.9742	1.0559	-18.92	94.71%
-10372.4	-10372.4	0.0000	0.0000	0.00	0.00%

Angle 90° Audit Summary:
Average Deviation: 4.72% | Maximum Divergence: 94.71%

Angle 135° Detailed Comparison (100 Points)

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
0.0	0.0	623.4480	681.5837	58.14	9.32%
-104.8	-104.8	641.8473	704.0364	62.19	9.69%
-209.5	-209.5	660.1622	726.2435	66.08	10.01%
-314.3	-314.3	678.3083	748.1857	69.88	10.30%
-419.1	-419.1	696.2014	769.3505	73.15	10.51%
-523.9	-523.9	713.7570	789.4632	75.71	10.61%
-628.6	-628.6	730.8907	808.0099	77.12	10.55%
-733.4	-733.4	747.5182	825.2185	77.70	10.39%
-838.2	-838.2	763.5552	841.4220	77.87	10.20%
-942.9	-942.9	778.9172	857.3692	78.45	10.07%
-1047.7	-1047.7	793.5198	873.0592	79.54	10.02%
-1152.5	-1152.5	807.2789	888.4369	81.16	10.05%
-1257.3	-1257.3	820.1296	902.8501	82.72	10.09%
-1362.0	-1362.0	832.0870	917.2632	85.18	10.24%
-1466.8	-1466.8	843.1854	929.0251	85.84	10.18%
-1571.6	-1571.6	853.4597	939.8324	86.37	10.12%
-1676.3	-1676.3	862.9442	950.6395	87.70	10.16%
-1781.1	-1781.1	871.6737	960.7314	89.06	10.22%
-1885.9	-1885.9	879.6828	970.5185	90.84	10.33%
-1990.7	-1990.7	887.0060	980.3056	93.30	10.52%
-2095.4	-2095.4	893.6780	989.1130	95.44	10.68%
-2200.2	-2200.2	899.7334	995.8938	96.16	10.69%
-2305.0	-2305.0	905.2067	1002.6746	97.47	10.77%
-2409.7	-2409.7	910.1280	1009.4553	99.33	10.91%
-2514.5	-2514.5	914.5087	1015.0216	100.51	10.99%
-2619.3	-2619.3	918.3556	1019.2551	100.90	10.99%
-2724.1	-2724.1	921.6757	1023.4886	101.81	11.05%
-2828.8	-2828.8	924.4757	1027.7222	103.25	11.17%
-2933.6	-2933.6	926.7625	1031.8048	105.04	11.33%
-3038.4	-3038.4	928.5430	1032.5440	104.00	11.20%
-3143.1	-3143.1	929.8239	1033.2833	103.46	11.13%
-3247.9	-3247.9	930.6122	1034.0226	103.41	11.11%
-3352.7	-3352.7	930.9147	1034.7618	103.85	11.16%
-3457.5	-3457.5	930.7383	1035.5011	104.76	11.26%
-3562.2	-3562.2	930.0917	1033.7361	103.64	11.14%
-3667.0	-3667.0	928.9916	1031.5861	102.59	11.04%
-3771.8	-3771.8	927.4568	1029.4361	101.98	11.00%
-3876.5	-3876.5	925.5060	1027.2860	101.78	11.00%
-3981.3	-3981.3	923.1577	1025.1360	101.98	11.05%
-4086.1	-4086.1	920.4305	1022.7545	102.32	11.12%
-4190.9	-4190.9	917.3434	1017.5314	100.19	10.92%
-4295.6	-4295.6	913.9148	1012.3083	98.39	10.77%
-4400.4	-4400.4	910.1635	1007.0852	96.92	10.65%
-4505.2	-4505.2	906.1081	1001.8621	95.75	10.57%
-4609.9	-4609.9	901.7673	996.6390	94.87	10.52%
-4714.7	-4714.7	897.1533	991.4160	94.26	10.51%
-4819.5	-4819.5	892.2523	984.8192	92.57	10.37%
-4924.3	-4924.3	887.0439	977.0278	89.98	10.14%
-5029.0	-5029.0	881.5079	969.2365	87.73	9.95%
-5133.8	-5133.8	875.6240	961.4452	85.82	9.80%
-5238.6	-5238.6	869.3718	953.6538	84.28	9.69%

Ref P (kN)	Calc P (kN)	Ref M (kNm)	Calc M (kNm)	Delta M	Error %
-5343.3	-5343.3	862.7311	945.8625	83.13	9.64%
-5448.1	-5448.1	855.6815	938.0712	82.39	9.63%
-5552.9	-5552.9	848.2028	929.0213	80.82	9.53%
-5657.7	-5657.7	840.2747	917.8818	77.61	9.24%
-5762.4	-5762.4	831.8768	906.7424	74.87	9.00%
-5867.2	-5867.2	822.9908	895.6029	72.61	8.82%
-5972.0	-5972.0	813.6057	884.4635	70.86	8.71%
-6076.7	-6076.7	803.7124	873.3241	69.61	8.66%
-6181.5	-6181.5	793.3020	862.1846	68.88	8.68%
-6286.3	-6286.3	782.3654	851.0451	68.68	8.78%
-6391.1	-6391.1	770.8935	837.3038	66.41	8.61%
-6495.8	-6495.8	758.8774	822.6392	63.76	8.40%
-6600.6	-6600.6	746.3079	807.9747	61.67	8.26%
-6705.4	-6705.4	733.1761	793.3102	60.13	8.20%
-6810.1	-6810.1	719.4728	778.6455	59.17	8.22%
-6914.9	-6914.9	705.1891	763.9810	58.79	8.34%
-7019.7	-7019.7	690.3159	749.3165	59.00	8.55%
-7124.4	-7124.4	674.8431	734.6519	59.81	8.86%
-7229.2	-7229.2	658.7609	717.1500	58.39	8.86%
-7334.0	-7334.0	642.0594	698.4570	56.40	8.78%
-7438.8	-7438.8	624.7286	679.7637	55.04	8.81%
-7543.5	-7543.5	606.7585	661.0706	54.31	8.95%
-7648.3	-7648.3	588.1392	642.3776	54.24	9.22%
-7753.1	-7753.1	568.8607	623.6843	54.82	9.64%
-7857.8	-7857.8	548.9130	604.9912	56.08	10.22%
-7962.6	-7962.6	528.2863	586.2982	58.01	10.98%
7.4	7.4	506.9704	679.9849	173.01	34.13%
-8172.2	-8172.2	484.9729	541.5805	56.61	11.67%
-8276.9	-8276.9	462.3700	518.3497	55.98	12.11%
-8381.7	-8381.7	439.2554	495.1189	55.86	12.72%
-8486.5	-8486.5	415.7229	471.8878	56.16	13.51%
-8591.2	-8591.2	391.8661	448.6570	56.79	14.49%
-8696.0	-8696.0	367.7788	425.4259	57.65	15.67%
-8800.8	-8800.8	343.5545	400.9282	57.37	16.70%
-8905.6	-8905.6	319.2870	371.8890	52.60	16.47%
-9010.3	-9010.3	295.0700	342.8527	47.78	16.19%
-9115.1	-9115.1	270.9972	313.8197	42.82	15.80%
-9219.9	-9219.9	247.1621	284.7901	37.63	15.22%
-9324.6	-9324.6	223.6380	255.7648	32.13	14.37%
-9429.4	-9429.4	200.4153	226.7444	26.33	13.14%
-9534.2	-9534.2	177.4638	197.7302	20.27	11.42%
-9639.0	-9639.0	154.7534	168.3725	13.62	8.80%
-9743.7	-9743.7	132.2539	138.3895	6.14	4.64%
-9848.5	-9848.5	109.9353	107.8352	-2.10	1.91%
-9953.3	-9953.3	87.7674	76.8243	-10.94	12.47%
-10058.0	-10058.0	65.7201	38.7430	-26.98	41.05%
-10162.8	-10162.8	43.7632	5.3966	-38.37	87.67%
-10267.6	-10267.6	21.8665	1.0090	-20.86	95.39%
-10372.4	-10372.4	0.0000	0.0000	0.00	0.00%

Angle 135° Audit Summary:
Average Deviation: 12.57% | Maximum Divergence: 95.39%